

The Diaz modulus conjecture: a machine-checked spine and the shape of its obstruction

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Abstract

Diaz asked in 2004 whether a non-zero complex number can be a logarithm of an algebraic number and have algebraic modulus. We give a short formal development of the question. All but one of the numbered statements below have been checked by machine, and the exception is flagged where it appears; Appendix A records where each proof lives, so that any claim here can be verified rather than taken.

The results divide in two. The first part is a spine: the strong four exponentials conjecture settles the question outright, by a single instantiation; a statement it implies, which we call (S) , settles the part of it that the elementary theory reduces to; and the value that a counterexample to (S) would exhibit cannot be a root of unity. The second part is negative, and is the reason for the paper. We show that one natural line of attack — assembling the certified data of a hypothetical counterexample into a matrix of logarithms with vanishing determinant — provably fails on one branch, that a second branch is circular in a precise sense, and that the remaining branch terminates in a theorem of 1973–74. When this note was first written that theorem had never been formalised; it has been since, in the form the branch needs, and the case it governs is proved with it. The natural interpolation route is blocked as well: no first-order operator with algebraic coefficients has algebraic values there, and the interpolation determinant does not see the candidate’s arithmetic at all. What is left is two statements, both open mathematics rather than formalisation, and we say exactly which. A third, which the development had carried until now, turns out not to be needed, and it and the statement it pairs with cannot both be false. Along the way the transcendence theorems now available formally yield a few unconditional results about classical constants, among them that at least one of e^{π^2} and $e^{i\pi^3}$ is transcendental. The smallest open instance of the conjecture, $\sqrt{(\log 2)^2 + \pi^2}$, is paired with $2^{i \log 2/\pi}$: at least one of the two is transcendental.

1 Setting

Write $\overline{\mathbb{Q}}$ for the algebraic numbers,

$$\mathcal{L} = \{\lambda \in \mathbb{C} : e^\lambda \in \overline{\mathbb{Q}}\}$$

for the logarithms of algebraic numbers, and $\tilde{\mathcal{L}}$ for the $\overline{\mathbb{Q}}$ -vector subspace of $\mathbb{C}$ spanned by 1 together with $\mathcal{L}$. Complex conjugation is written $\bar{}$, and $\overline{e^u} = e^{\bar{u}}$, so $\mathcal{L}$ is stable under it.

Conjecture 1.1 (Diaz, 2004 [1]). *For every $u \in \mathbb{C}$ with $u \neq 0$ and $|u| \in \overline{\mathbb{Q}}$, the number e^u is transcendental.*

We call u a *candidate* when it violates this: $u \neq 0$, $|u| \in \overline{\mathbb{Q}}$, and $e^u \in \overline{\mathbb{Q}}$. A candidate carries exactly four pieces of certified data,

$$u \in \mathcal{L}, \quad \bar{u} \in \mathcal{L}, \quad 2\pi i \in \mathcal{L}, \quad u\bar{u} = |u|^2 \in \overline{\mathbb{Q}},$$

and the last of these is a *quadratic* relation between logarithms. That is the whole source of the difficulty: the linear theory of logarithms — Baker’s theorem and its relatives — says nothing about products, and the theory that does speak about products of two logarithms is the four exponentials circle. Section 2 makes this precise, and everything after it is an attempt to ask for less.

Two classical inputs are used without proof. *Hermite–Lindemann*: for algebraic $a \neq 0$, e^a is transcendental. *Transcendence of π* , and hence of π^2 . Three further classical theorems enter in the results added in version 1.1, and all three are machine-checked: the Gel’fond–Schneider theorem, in the formalisation of Karatarakis and Wiedijk [3]; the six exponentials theorem; and the four exponentials theorem in transcendence degree one (Section 5).

Conjecture 1.2 (Strong four exponentials). *Let x_1, x_2 be linearly independent over $\overline{\mathbb{Q}}$ and let y_1, y_2 be linearly independent over $\overline{\mathbb{Q}}$. Then at least one of the four products $x_i y_j$ lies outside $\tilde{\mathcal{L}}$.*

This is Waldschmidt’s strengthening of the four exponentials conjecture, in which linear independence is taken over $\overline{\mathbb{Q}}$ rather than over $\mathbb{Q}$. Both are open.

2 The ceiling

Theorem 2.1. *Conjecture 1.2 implies Conjecture 1.1.*

Proof. Let $u \neq 0$ with $|u| \in \overline{\mathbb{Q}}$, and suppose $e^u \in \overline{\mathbb{Q}}$.

If u is algebraic, Hermite–Lindemann gives that e^u is transcendental, a contradiction. So u is transcendental, and so is $\bar{u}$, since the conjugate of an algebraic number is algebraic. Consequently $(1, u)$ and $(1, \bar{u})$ are each linearly independent over $\overline{\mathbb{Q}}$: a relation $a + bu = 0$ with $a, b \in \overline{\mathbb{Q}}$ not both zero would force either $b \neq 0$ and $u = -a/b \in \overline{\mathbb{Q}}$, or $b = 0$ and $a = 0$.

Apply Conjecture 1.2 to $x = (1, u)$ and $y = (1, \bar{u})$. The four products are

$$1, \quad \bar{u}, \quad u, \quad u\bar{u} = |u|^2,$$

and all four lie in $\tilde{\mathcal{L}}$: the first by definition of $\tilde{\mathcal{L}}$; the second and third because $e^{\bar{u}} = \overline{e^u}$ and e^u are algebraic, so $u, \bar{u} \in \mathcal{L}$; and the fourth because $|u|^2$ is algebraic and $\overline{\mathbb{Q}} \cdot 1 \subseteq \tilde{\mathcal{L}}$. This contradicts Conjecture 1.2. $\square$

Remark 2.2. Schanuel’s conjecture also suffices, by a different route. If u and $\bar{u}$ are linearly independent over $\mathbb{Q}$, Schanuel at $n = 2$ gives $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}, e^u, e^{\bar{u}}) \geq 2$, hence $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}) \geq 2$ once the two exponentials are algebraic; but $u\bar{u} \in \overline{\mathbb{Q}}$ is a non-trivial algebraic relation, so that degree is at most 1. If instead $u = q\bar{u}$ with $q \in \mathbb{Q}$, then $|q| = 1$, so $q = \pm 1$ and $u^2 = \pm |u|^2 \in \overline{\mathbb{Q}}$, making u algebraic; Hermite–Lindemann finishes. Schanuel at $n = 1$ is itself Hermite–Lindemann, so no separate input is needed.

Theorem 2.1 fixes the ceiling of the whole subject. Any decomposition of Conjecture 1.1 into cases produces statements that are consequences of Conjecture 1.2, and the useful question is not whether a case follows from it — all of them do — but whether a case follows from something *weaker*. The next section exhibits one such statement.

3 Quantisation and the period lattice

This section is self-contained: apart from the transcendence of π and of π^2 it uses no arithmetic input at all. It ends by collapsing an entire branch of the problem to a single sentence about a single real variable.

Throughout, u is off both coordinate axes, meaning $\text{Re } u \neq 0$ and $\text{Im } u \neq 0$.

Theorem 3.1 (Quantisation). *Let u be off both axes with e^u real. Then $\operatorname{Im} u \in \pi\mathbb{Z} \setminus \{0\}$, and $|u|^2 > \pi^2$.*

Proof. Write $u = t + i\theta$. Then $e^u = e^t(\cos \theta + i \sin \theta)$, so e^u real forces $e^t \sin \theta = 0$, and $e^t \neq 0$ gives $\sin \theta = 0$, that is $\theta = k\pi$ for some $k \in \mathbb{Z}$. Since $\operatorname{Im} u \neq 0$ we have $k \neq 0$. Then $|u|^2 = t^2 + k^2\pi^2 \geq t^2 + \pi^2 > \pi^2$, using $t \neq 0$. $\square$

The lower bound is the only effective statement in this development, and it sharpens: if the ratio $e^u/\overline{e^u}$ has finite order m , the same argument applied to u/m gives $|u|^2 > \pi^2/m^2$. What that sharpening is worth is settled by the next statement, which says the whole family carries exactly one bit of information.

Proposition 3.2. *For u off the axes, the quantisation conclusion holds across the entire orbit $\{qu : q \in \mathbb{Q}^\times\}$ if and only if $\operatorname{Re} u \neq 0$.*

No algebraicity is used. The point is negative: a predicate that looks like a family of increasingly precise constraints is equivalent to the hypothesis one already has, so no split of the problem along m can be honest.

Theorem 3.3 (Fibres). *Let $\alpha \in \overline{\mathbb{Q}}^\times$. At most two of the logarithms of α have algebraic modulus.*

Proof. The logarithms of α are $u + 2\pi in$, $n \in \mathbb{Z}$, for any one of them u . Writing $\theta = \operatorname{Im} u$,

$$|u + 2\pi in|^2 = |u|^2 + 4\pi n\theta + 4\pi^2 n^2.$$

Suppose n_1, n_2, n_3 are distinct and all three give an algebraic modulus, hence an algebraic squared modulus. Subtracting the n_2 equation from the n_1 one and dividing by $4(n_1 - n_2) \neq 0$ gives $\pi\theta + \pi^2(n_1 + n_2) \in \overline{\mathbb{Q}}$, and likewise $\pi\theta + \pi^2(n_1 + n_3) \in \overline{\mathbb{Q}}$. Subtracting these, $\pi^2(n_2 - n_3) \in \overline{\mathbb{Q}}$ with $n_2 \neq n_3$, so $\pi^2 \in \overline{\mathbb{Q}}$ — false. $\square$

Corollary 3.4. *If two distinct logarithms of α have algebraic modulus, say u and $u + 2\pi in$ with $n \neq 0$, then $\pi(\operatorname{Im} u + n\pi) \in \overline{\mathbb{Q}}$.*

Proof. Immediate from the displayed identity: the difference of the two squared moduli is $4\pi n\theta + 4\pi^2 n^2$, algebraic, and dividing by $4n$ gives $\pi\theta + \pi^2 n = \pi(\theta + n\pi) \in \overline{\mathbb{Q}}$. $\square$

Corollary 3.4 is worth reading twice. The condition it produces was introduced elsewhere in this subject as a technical case distinction; what it says is that the fibre over e^u contains a *second* point of algebraic modulus. Theorem 3.3 bounds how often that can happen, and eliminating $\operatorname{Im} u$ between the two displayed relations returns a quartic relation in π — which is the reason the class it cuts out has transcendence degree one over $\mathbb{Q}$. The period lattice contributes one bit and no more.

One sentence

Theorem 3.5. *The following are equivalent.*

- (i) *For every $u \neq 0$ off both axes with $|u| \in \overline{\mathbb{Q}}$ and e^u real, e^u is transcendental.*
- (ii) *For every real $t \neq 0$ with $e^t \in \overline{\mathbb{Q}}$, the number $t^2 + \pi^2$ is transcendental.*

Proof. Assume (ii) and let u be as in (i) with $e^u \in \overline{\mathbb{Q}}$. By Theorem 3.1, $u = t + ik\pi$ with $t = \operatorname{Re} u \neq 0$ and $k \in \mathbb{Z} \setminus \{0\}$. Put $v = u/k$. Then e^v is a k -th root of e^u , hence algebraic; $\operatorname{Im} v = \pi$; $\operatorname{Re} v = t/k \neq 0$; and $|v|^2 = (t/k)^2 + \pi^2$, which is algebraic because $|u|^2$ is and k is rational. Also e^v is real up to sign, so $e^{t/k} = \pm e^v$ is algebraic. Applying (ii) at t/k makes $(t/k)^2 + \pi^2$ transcendental, contradicting the previous sentence.

Conversely assume (i) and let $t \neq 0$ be real with $e^t \in \overline{\mathbb{Q}}$ and suppose $t^2 + \pi^2 \in \overline{\mathbb{Q}}$. Set $u = t + i\pi$. Then $e^u = -e^t$ is real and algebraic, $|u|^2 = t^2 + \pi^2$ is algebraic, and u is off both axes. This contradicts (i). $\square$

The k in that proof carries no content, which is the substance of the theorem: an entire branch of the problem, quantified over two real parameters and an integer, is one assertion about one real variable. Its smallest instance is a question one can state without any of the preceding apparatus.

Open question. Is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?

Nothing in this development answers it. Section 2 shows the strong four exponentials conjecture does; Section 4 explains why the available unconditional theory does not reach it. What that theory does give is a partner that must be transcendental if the answer is yes.

Theorem 3.6. *Let $t \neq 0$ be real with $e^t \in \overline{\mathbb{Q}}$, and suppose $\rho = t^2 + \pi^2$ is algebraic. Then $e^{it^2/\pi}$, $e^{\pi^2/t}$ and $e^{\rho/(i\pi)}$ are transcendental.*

Proof. The hypothesis makes t algebraic over $\mathbb{Q}(\pi)$, since $t^2 = \rho - \pi^2$; this is its only use. With transcendence degree one available, Proposition 4.4 applies to $(w, z) = (i\pi, t/(i\pi))$, where $e^w = -1$ and $wz = t$, so $wz^2 = -it^2/\pi \notin \mathcal{L}$; and to $(w, z) = (t, i\pi/t)$, where $wz = i\pi$, so $wz^2 = -\pi^2/t \notin \mathcal{L}$. Finally $\rho/(i\pi) = -it^2/\pi - i\pi$, so $e^{\rho/(i\pi)} = -e^{-it^2/\pi}$. $\square$

Corollary 3.7. *At least one of $\sqrt{(\log 2)^2 + \pi^2}$ and $2^{i \log 2/\pi} = e^{i(\log 2)^2/\pi}$ is transcendental.*

The third number in Theorem 3.6 couples two open statements. The real half of (S) , below, asks that $e^{\gamma/(i\pi)}$ be transcendental for real algebraic $\gamma \neq 0$; the theorem proves it at every $\gamma = t^2 + \pi^2$ that the present question would make algebraic. The two remaining open statements of Section 7 cannot fail at the same number.

A second point in the fibre, and the statement (S)

Corollary 3.4 says what it takes for the fibre over e^u to contain a second point of algebraic modulus. On the branch where the angle is irrational, $\operatorname{Im} u \notin \pi\mathbb{Q}$, that condition does more than constrain u : it manufactures a new logarithm.

Proposition 3.8. *Let $u \in \mathcal{L}$ with $\operatorname{Im} u \notin \pi\mathbb{Q}$, and suppose $\pi(\operatorname{Im} u + r\pi) \in \overline{\mathbb{Q}}$ for some rational $r \neq 0$. Then there is a non-zero algebraic γ with $e^{\gamma/(i\pi)}$ algebraic.*

Proof. $\mathcal{L}$ is a $\mathbb{Q}$ -vector space — $e^{(a/b)\lambda}$ is a root of $X^b - (e^\lambda)^a$ — and it contains u , $\bar{u}$ and $2\pi i$. So it contains $\lambda = \frac{1}{2}(u - \bar{u}) + \frac{r}{2}(2\pi i) = i(\operatorname{Im} u + r\pi)$. Put $\gamma = i\pi\lambda = -\pi(\operatorname{Im} u + r\pi)$. It is algebraic by hypothesis, and non-zero because $\operatorname{Im} u = -r\pi$ is excluded; and $\gamma/(i\pi) = \lambda \in \mathcal{L}$. $\square$

So on this branch a candidate would exhibit a non-zero algebraic multiple of $1/(i\pi)$ inside $\mathcal{L}$. That is the statement the branch reduces to.

Conjecture 3.9 $((S))$. *For every non-zero algebraic γ , the number $e^{\gamma/(i\pi)}$ is transcendental. Equivalently, no non-zero algebraic multiple of $1/(i\pi)$ lies in $\mathcal{L}$.*

Theorem 3.10. *Conjecture 1.2 implies Conjecture 3.9.*

Proof. Suppose $\gamma \neq 0$ is algebraic and $\lambda = \gamma/(i\pi) \in \mathcal{L}$. Apply Conjecture 1.2 to $x = (1, \lambda)$ and $y = (1, i\pi)$. The four products are 1, $i\pi$, λ and γ , all in $\tilde{\mathcal{L}}$. Both pairs are $\mathbb{Q}$ -linearly independent, since $i\pi$ is transcendental and so is λ , whose algebraicity would make $i\pi = \gamma/\lambda$ algebraic. This contradicts Conjecture 1.2. $\square$

Whether (S) is strictly weaker than Conjecture 1.2 is not known; what is known is that it asks for much less, one number against a whole configuration. Its hypothetical counterexamples are also more constrained than the value $e^{\gamma/(i\pi)}$ suggests.

Proposition 3.11. *For every non-zero algebraic γ , the number $e^{\gamma/(i\pi)}$ is not a root of unity.*

Proof. If $e^{\gamma/(i\pi)} = e^{2\pi i k/n}$, then $\gamma/(i\pi) = 2\pi i(k/n + m)$ for some $m \in \mathbb{Z}$, so $\gamma = -2\pi^2(k/n + m)$. As $\gamma \neq 0$, the rational $k/n + m$ is non-zero and π^2 is algebraic — false. $\square$

The statement also splits, and not by a case distinction. The set $S_0 = \{\gamma \in \overline{\mathbb{Q}} : \gamma/(i\pi) \in \mathcal{L}\}$ is a $\mathbb{Q}$ -vector space, and it is stable under conjugation because $\overline{i\pi} = -i\pi$ gives $\bar{\gamma}/(i\pi) = -\gamma/(i\pi)$. So with γ it contains $\operatorname{Re} \gamma$ and $i \operatorname{Im} \gamma$, and (S) is equivalent to its two axis halves: for real $\gamma \neq 0$, the number $e^{-i\gamma/\pi}$, which lies on the unit circle and is not a root of unity, is transcendental; and for $\gamma = i\beta$ with β real, $e^{\beta/\pi}$ is transcendental — that is, $\pi \neq \beta/\log \alpha$ for real algebraic $\beta \neq 0$ and algebraic $\alpha > 0$, $\alpha \neq 1$. The reduction is machine-checked, and both halves are open. They cannot both fail.

Theorem 3.12. (a) *If $\gamma_1, \gamma_2 \in S_0$ and $\gamma_2 \neq 0$, then $\gamma_1/\gamma_2 \in \mathbb{Q}$.*

(b) *Every element of S_0 is real or purely imaginary.*

(c) *At least one of the two halves of (S) holds.*

Proof. (a) The numbers $\lambda_j = \gamma_j/(i\pi)$ lie in $\mathcal{L}$, $\lambda_2 \neq 0$, and their ratio γ_1/γ_2 is algebraic. If it were irrational, Gel'fond–Schneider would make $e^{\lambda_1} = e^{(\gamma_1/\gamma_2)\lambda_2}$ transcendental. (b) By the conjugation stability of S_0 and (a), $\bar{\gamma} = q\gamma$ with $q \in \mathbb{Q}$; comparing real and imaginary parts, either $\operatorname{Re} \gamma = 0$, or $q = 1$ and $\operatorname{Im} \gamma = 0$. (c) A non-zero real element and a non-zero purely imaginary one would be rationally proportional by (a). $\square$

So S_0 is either $\{0\}$ or a single rational line on one axis, and (S) can fail, if at all, in one direction only. The imaginary half has a concrete reading: its exceptions are the $\beta = \pi \log \alpha$ that happen to be algebraic, and by (a) all such $\log \alpha$ are rationally proportional. For instance, at least one of $\pi \log 2$ and $\pi \log 3$ is transcendental, although neither is known to be.

Each single instance of (S) also has a partner. For every algebraic $\gamma \neq 0$, at least one of $e^{i\gamma/\pi}$ and $e^{i\pi^3/\gamma}$ is transcendental, and at least one of $e^{i\gamma/\pi}$ and $e^{i\gamma^2/\pi^3}$ is (Proposition 4.4 with $(w, z) = (\gamma/(i\pi), -\pi^2/\gamma)$ and $(i\pi, -\gamma/\pi^2)$). At $\gamma = 1$: at least one of $e^{i/\pi}$ and $e^{i\pi^3}$ is transcendental.

Which direction matters for Conjecture 1.1 is settled by looking at what the branch produces. In Proposition 3.8 the number $\gamma = -\pi(\operatorname{Im} u + r\pi)$ is real. The formal decomposition reaches (S) along one other branch, where $\pi \operatorname{Im} u$ itself is algebraic, and there $\gamma = -\pi \operatorname{Im} u$ is real as well. Both branches have been reduced, by machine-checked reductions, to the real half alone. The imaginary half is therefore not needed anywhere in the development (Section 7).

4 The precise open boundary

Attach to a candidate u , with $\rho = u\bar{u}$ and $r = \sqrt{\rho}$, the matrix

$$H(u, r) = \begin{pmatrix} u & r \\ r & \bar{u} \end{pmatrix}, \quad \det H(u, r) = u\bar{u} - r^2 = 0.$$

Its rows are dependent over $\mathbb{C}$. The next statement says they are dependent in no weaker sense, and it is the sharpest way to see where this problem sits.

Theorem 4.1 (No vanishing coefficient). *Let $K \subseteq \mathbb{C}$ be a subfield with $r \in K$, $r \neq 0$, and let u be transcendental over K with $u\bar{u} = r^2$. Then for all non-zero $w, v \in K^2$,*

$$w^\top H(u, r) v \neq 0.$$

Proof. Expanding, $w^\top H v = w_0(uv_0 + rv_1) + w_1(rv_0 + \bar{u}v_1)$. Using $\bar{u} = r^2/u$ and clearing the denominator $u \neq 0$, the vanishing of this expression is a quadratic relation for u over K with coefficients built from w, v, r ; its leading coefficient is w_0v_0 and its constant term

is $r^2 w_1 v_1$. A vanishing of the expression makes u a root of a polynomial over K , so by transcendence that polynomial is identically zero: $w_0 v_0 = 0$, $r^2 w_1 v_1 = 0$, and the middle coefficient $r(w_0 v_1 + w_1 v_0) = 0$. With $r \neq 0$ the first two give $w_0 v_0 = w_1 v_1 = 0$, and combining with the third forces $w = 0$ or $v = 0$. $\square$

At $K = \overline{\mathbb{Q}}$ the transcendence hypothesis is exactly what a candidate supplies: an algebraic u with e^u algebraic contradicts Hermite–Lindemann, so a candidate is transcendental.

So $H(u, r)$ is singular over $\mathbb{C}$ while every coefficient extracted from it over $\overline{\mathbb{Q}}$ is non-zero. A candidate hides behind exactly this discrepancy: rank drops over $\mathbb{C}$ and nothing detects it over $\overline{\mathbb{Q}}$.

One might hope this configuration is an artefact of a particular choice of matrix. It is not.

Theorem 4.2 (Uniqueness of the obstruction). *Let $K \subseteq \mathbb{C}$ be a subfield, u transcendental over K with $u \neq 0$ and $u\bar{u} \in K$, and let A, B, C be 2×2 matrices over K with $\det(A + uB + \bar{u}C) = 0$. Then there is $c \in K$ with*

$$\det(xA + yB + zC) = c(yz - (u\bar{u})x^2) \quad \text{for all } x, y, z \in \mathbb{C}.$$

Every singular pencil over the candidate's hull is therefore the same object: its determinant is a K -multiple of the single quadratic form $yz - \rho x^2$. Concretely, $H(u, r)$ factors as

$$\begin{pmatrix} 0 & r \\ r & 0 \end{pmatrix} (I + uB_0 + \bar{u}C_0), \quad B_0^2 = C_0^2 = 0, \quad \text{tr}(B_0 C_0) = \rho^{-1},$$

so the whole family is a nilpotent pencil pinned by one trace. There is no second configuration to look for, and no gain in enlarging the search.

Descent to a conic, and what is missing

Theorem 4.2 sends the problem to the affine conic

$$X_\rho : XY = \rho,$$

on which a candidate is the point $(u, \bar{u}) \in \mathcal{L}^2$. This makes the boundary explicit, and it also makes a tempting route collapse.

Proposition 4.3. *Fix $\rho \neq 0$ and let $L \subseteq \mathbb{C}$ be any set. Suppose every point of $X_\rho \cap L^2$ lies in a $\mathbb{Q}$ -rational vector subspace of $\mathbb{C}^2$ contained in X_ρ . Then $X_\rho \cap L^2 = \emptyset$.*

Proof. A vector subspace contains the origin, and $0 \cdot 0 = 0 \neq \rho$, so no vector subspace is contained in X_ρ . The hypothesis can only be satisfied vacuously. $\square$

The statement is trivial and the reason to record it is what it exposes when read backwards. Roy's conjecture in the shape just used — points of a variety with coordinates in $\mathcal{L}$ lie on rational subspaces contained in it — is available for *homogeneous* quadrics, where there are non-trivial subspaces to be placed on. The affine conic offers none, so the same hypothesis becomes an emptiness statement rather than a structural one, and the gap between the two is not a technicality but the location of the whole difficulty.

Stated in the form that is actually available: Proposition 12.25 of Waldschmidt's book, for arbitrary $k \subseteq K$ and a k -subspace $\mathcal{E}$ of K , gives the conclusion for a quadric defined over k provided every non-zero matrix M over $\mathcal{E}$ satisfies

$$\text{rank}(M) > \frac{1}{2} r_{\text{str}}(M),$$

the structural rank taken over k . Two things separate that from Diaz's conjecture, and they are of different kinds.

- (a) **The rank inequality.** Over $(\mathbb{Q}, \mathbb{C}, \mathcal{L})$ it is supplied at transcendence degree one by Theorem 15.30 of the same book — which is why the transcendence-degree-one statements in this area are unconditional, and equally why they are not new. Over $(\overline{\mathbb{Q}}, \mathbb{C}, \tilde{\mathcal{L}})$, the coefficient field this problem requires, only the non-strict bound of Corollary 12.22 is known. This is being worked on; the Structural Rank Conjecture of Dasgupta and Kakde asserts something stronger.
- (b) **Homogeneity.** Proposition 12.25 covers homogeneous quadrics; X_ρ is affine. Waldschmidt makes the point himself, observing that the inhomogeneous extension would give the transcendence of e^{λ^2} , and that e^{π^2} remains open. This is not being worked on.

The distinction is worth keeping. An expectation that the subject will drift towards Diaz's conjecture is reasonable for (a) and unsupported for (b), and it is (b) that Theorem 3.5 runs into: $\sqrt{(\log 2)^2 + \pi^2}$ asks about a *sum*, and every available tool is homogeneous.

What the homogeneous theory does give at e^{π^2} is worth recording, because it is unconditional and machine-checked. The four exponentials theorem in transcendence degree one (Section 5) applies to the simplest singular matrix whose entries are tied by one ratio.

Proposition 4.4. *Let $w \neq 0$ and $z \notin \mathbb{Q}$ with $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(w, z) \leq 1$. Then w , wz and wz^2 are not all in $\mathcal{L}$.*

Proof. Otherwise the matrix $\begin{pmatrix} w & wz \\ wz & wz^2 \end{pmatrix}$ has vanishing determinant and entries in $\mathcal{L}$ generating transcendence degree at most one, so its rows or columns satisfy a non-trivial rational relation $pw + qwz = 0$; this forces $z = -p/q \in \mathbb{Q}$. $\square$

Corollary 4.5. *At least one of e^{π^2} and $e^{i\pi^3}$ is transcendental.*

Proof. Take $w = i\pi$ and $z = -i\pi$: then $e^w = -1$, $wz = \pi^2$ and $wz^2 = -i\pi^3$. $\square$

This is as close to e^{π^2} as the homogeneous theory comes.

5 Two barriers, and one branch that returns

Section 4 located the difficulty. This section records what is known about methods rather than about the problem: two results saying that whole classes of approach cannot work, and one observation about decomposition itself.

The candidate is invisible to algebraic coefficients

Theorem 5.1 (Indistinguishability). *Let $u \neq 0$ with e^u algebraic, and let $r \in \overline{\mathbb{Q}}$ be real and non-zero with $u\bar{u} = r^2$. Then there exist a transcendental t with $t\bar{t} = r^2$ and a ring homomorphism $\Phi : \mathbb{C} \rightarrow \mathbb{C}$ fixing $\overline{\mathbb{Q}}$ pointwise, carrying u to t , commuting with conjugation on $\overline{\mathbb{Q}}(u)$, such that for every matrix M over $\overline{\mathbb{Q}}(u)$ and all vectors w, v over $\overline{\mathbb{Q}}$,*

$$w^T M v = 0 \iff w^T \Phi(M) v = 0.$$

The point t can be any transcendental point of the same circle, for instance re^i . It can also be chosen with e^t transcendental, since the circle is uncountable while $\overline{\mathbb{Q}}$ and $\mathcal{L}$ are countable. Whether re^i itself is a candidate is an instance of the conjecture, so version 1.2's " re^i will do" for a non-candidate was too quick. So no vanishing statement with algebraic coefficients, formulated over the candidate's own hull, separates a hypothetical candidate from a point that is certainly not one. Any argument that would refute a candidate by exhibiting such a vanishing relation refutes nothing.

That the intertwining with conjugation comes for free is worth a line: since $\bar{u} = \rho/u$, conjugation is a rational function of the generator on $\overline{\mathbb{Q}}(u)$, so any Φ matching generators commutes with it automatically. The symmetry is not arranged; it is forced.

The threshold is a dimension

Theorem 5.2 (Dimension count). *Let u be a candidate. Then $1, u, \bar{u}$ all lie in $\tilde{\mathcal{L}}$ and, granting Hermite–Lindemann, they span an $\overline{\mathbb{Q}}$ -subspace of dimension exactly 3. There exist $\overline{\mathbb{Q}}$ -independent pairs $x = (x_1, x_2)$ and $y = (y_1, y_2)$ with all four products $x_i y_j$ in that span; and for no $\overline{\mathbb{Q}}$ -independent x of length 2 and y of length 3 do all six products $x_i y_j$ lie in it.*

The certificate a candidate offers is three-dimensional. A 2×2 configuration needs four products and fits; a 2×3 configuration needs six and cannot. That is the whole reason the strong four exponentials conjecture reaches this problem while the six exponentials *theorem* — which is proved, and which would otherwise be the natural instrument — does not. The obstruction is not depth but dimension, and no sharpening of six exponentials that stays inside a three-dimensional certificate space can help. Where there is room the theorem works as expected: with $x = (1, \pi)$ and $y = (\log 2, \log 3, \log 5)$ it shows that at least one of $2^\pi, 3^\pi, 5^\pi$ is transcendental; and for $\mathbb{Q}$ -independent $\lambda, \mu \in \mathcal{L}$, at least one of $e^{\lambda^2/\mu}$ and $e^{\mu^2/\lambda}$ is, so that one of $2^{\log_3 2}$ and $3^{\log_2 3}$ is. The room can also come from a second candidate: Section 6 shows that two candidates of commensurable moduli give the theorem the fourth dimension a single one lacks.

The four exponentials conjecture sees nothing homogeneous

Theorem 5.2 works over $\overline{\mathbb{Q}}$ and counts dimensions. Over $\mathbb{Q}$, the field of the four exponentials conjecture itself, dimension is the wrong invariant: $x = y = (1, \sqrt{2})$ is a 2×2 configuration inside a two-dimensional space. The right invariant is the absence of homogeneous relations. Call a candidate *generic* if u and π are algebraically independent (over $\overline{\mathbb{Q}}$, equivalently over $\mathbb{Q}$); otherwise u is algebraic over $\mathbb{Q}(\pi)$.

Theorem 5.3. *Let $u \neq 0$ with $|u|^2 \in \overline{\mathbb{Q}}$, and suppose u and π are algebraically independent over $\overline{\mathbb{Q}}$.*

- (a) *No non-zero rational quadratic form vanishes at $(u, \bar{u}, i\pi)$.*
- (b) *Every singular 2×2 matrix with entries in $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has $\mathbb{Q}$ -dependent rows or $\mathbb{Q}$ -dependent columns.*

Proof. (a) Substitute $\bar{u} = \rho/u$ and multiply by u^2 . The result is a polynomial in u and $i\pi$ with the six distinct monomials $u^4, u^3 i\pi, u^2 (i\pi)^2, u^2, u i\pi, 1$ and coefficients the symmetrised coefficients of the form, three of them multiplied by ρ or ρ^2 . By independence all vanish. (b) The determinant is a rational quadratic form in $(u, \bar{u}, i\pi)$, hence identically zero by (a). A 2×2 matrix of linear forms with identically vanishing determinant has dependent rows or columns, by unique factorisation in the polynomial ring. $\square$

Nothing here uses e^u : the statement is about any point of the circle that is generic in this sense. For a candidate it says that the four exponentials *conjecture*, fed with the logarithms the candidate certifies, has no instance at all. The non-generic candidates are no better targets, for the opposite reason.

Proposition 5.4. *Let u be a candidate algebraic over $\mathbb{Q}(\pi)$. Then every singular 2×2 matrix with entries in $\mathbb{Q}u + \mathbb{Q}\bar{u} + \mathbb{Q}i\pi$ has $\mathbb{Q}$ -dependent rows or $\mathbb{Q}$ -dependent columns.*

Proof. A zero entry forces a zero row or column. Otherwise all entries lie in $\mathcal{L}$ and are algebraic over $\mathbb{Q}(\pi)$, and the four exponentials theorem in transcendence degree one applies. $\square$

So for every candidate, the four exponentials conjecture restricted to its certified logarithms is already a theorem: by algebra in the generic case, by the 1973–74 theorem otherwise. The same holds for an exception $\lambda = \gamma/(i\pi)$ to (S) : every singular 2×2 matrix over $\mathbb{Q}\lambda + \mathbb{Q}i\pi$ has dependent rows or columns, because $\lambda/(i\pi) = -\gamma/\pi^2$ is transcendental. What does detect a

counterexample needs a constant term. The configurations of Theorems 2.1 and 3.10 have one. In the generic case, one can show that the period $i\pi$ never enters a configuration that detects a candidate (this is not formalised). Non-generic candidates can carry homogeneous relations. For example $\operatorname{Re}(u^2) = \pi^2$ is satisfied by points of every circle of algebraic radius $r > \pi$. But such a relation is never the determinant of a 2×2 matrix over the certified logarithms unless the four exponentials theorem in transcendence degree one already excludes it.

A decomposition that returns to its start

Splitting the conjecture into cases produces statements that are formally weaker, and it is tempting to read a long chain of such splits as progress. It need not be.

Proposition 5.5. *Let (NT) be the assertion: for every u with e^u and $e^{\bar{u}}$ algebraic, $\operatorname{Re} u \neq 0$, $\operatorname{Im} u \neq 0$, and $\operatorname{Re} u$, $\operatorname{Im} u$, $\operatorname{Re} u / \operatorname{Im} u$ all transcendental, the modulus $|u|$ is transcendental. Then, granting Hermite–Lindemann, (NT) is equivalent to Conjecture 1.1.*

Proof. Assume (NT) and let $u \neq 0$ have $|u| \in \overline{\mathbb{Q}}$ and $e^u \in \overline{\mathbb{Q}}$. Then $e^{\bar{u}} = \overline{e^u}$ is algebraic. If any of $\operatorname{Re} u$, $\operatorname{Im} u$, $\operatorname{Re} u / \operatorname{Im} u$ is algebraic, or if either of $\operatorname{Re} u$, $\operatorname{Im} u$ vanishes, then $|u|^2 \in \overline{\mathbb{Q}}$ forces u itself algebraic — in each case one coordinate is algebraic and the identity $|u|^2 = (\operatorname{Re} u)^2 + (\operatorname{Im} u)^2$ supplies the other — and Hermite–Lindemann applies. Otherwise (NT) makes $|u|$ transcendental, contradicting the hypothesis. The converse is immediate: a u witnessing the failure of (NT) has algebraic modulus and algebraic e^u . $\square$

(NT) is recorded formally and is open; the equivalence just proved is not itself formalised, and Appendix A marks it so.

(NT) is a natural-looking restriction of the conjecture, reached by imposing genericity on a conjugate pair, and it sits several case splits below the original statement. It is nonetheless the original statement. A decomposition can descend a long way and arrive back where it began, and only closing the cycle detects this; reading the chain forwards never does.

What is not available

Two standard instruments are inapplicable here for reasons that are structural rather than quantitative, and it is worth recording which.

Gel’fond–Schneider produces transcendence of α^β from an algebraic $\alpha \neq 0, 1$ and an algebraic irrational β . Applied to the configuration above it manufactures transcendence at a point unrelated to the one in question; the number whose transcendence is wanted is not of that shape. It is not idle elsewhere: it is what confines the exceptions to (S) to one rational line (Theorem 3.12).

Baker’s theorem bounds non-vanishing linear forms in logarithms from below. The relevant form here vanishes identically by hypothesis — that is what being a candidate means — so the bound has nothing to act on. Using it would require knowing in advance that the form is non-zero, which is the conclusion sought.

Finally, the transcendence-degree-one case of four exponentials, which does apply on part of the locus, is a theorem of 1973–74 (Brownawell; Waldschmidt), recorded as Theorem 1 of Roy–Waldschmidt [4]. When this note was first written it was available for citation and for nothing else, and the obstacle to a formal development looked like Philippon’s zero estimate together with the apparatus around it. That reading was wrong about the route. Waldschmidt’s own proof [6], with the toolbox of his 1971 paper [7], uses none of it: Siegel’s lemma, a Gel’fond-type transcendence criterion, the maximum principle, and a zero count for exponential polynomials that needs no separation hypothesis — the last being what removes Baker’s theorem from the 1973 argument. Along that route the theorem is now formalised, and the case of the branch that it governs is proved with it; see Appendix A. What remains on that branch is not the

1973–74 theorem but two statements about $1/\pi$, both open. The formal statement takes all four exponentials to be algebraic, which is more than the 1973 theorem assumes and exactly what this setting supplies.

Interpolation on the lattice sees nothing

The classical way to attack e^u and $e^{\bar{u}}$ together is to run the two-variable exponential system

$$F(z, w) = \exp(uz + \bar{u}w)$$

on the lattice $\mathbb{Z}^2$, where its values $F(m, n) = e^{mu}e^{n\bar{u}}$ are algebraic, and to feed an auxiliary function built from it to an interpolation or Schneider–Lang argument. Two elementary facts stand in the way.

Proposition 5.6 (No first-order arithmetic operator). *Let u be a candidate and $X = a\partial_z + b\partial_w$ with a, b polynomials with algebraic coefficients. If $(XF)(m, n)$ is algebraic for every $(m, n) \in \mathbb{Z}^2$, then $a = b = 0$.*

Proof. $(XF)(m, n) = (a(m, n)u + b(m, n)\bar{u})F(m, n)$, and $F(m, n)$ is a non-zero algebraic number, so $a(m, n)u + b(m, n)\bar{u}$ is algebraic. A candidate lies on neither axis — on an axis, $|u| \in \overline{\mathbb{Q}}$ would make u algebraic, against Hermite–Lindemann — so u and $\bar{u}$ are $\mathbb{Q}$ -linearly independent, as in the proof of Theorem 6.1, and Baker’s theorem in its inhomogeneous form gives $a(m, n) = b(m, n) = 0$. A polynomial in two variables vanishing on $\mathbb{Z}^2$ is zero. $\square$

The same holds for a, b rational functions with algebraic coefficients regular on $\mathbb{Z}^2$, by clearing denominators; the formal statement is the polynomial one. The consequence is that the criterion of Schneider–Lang [8, Theorem 4.1], which asks the derivatives of the functions to stay in a ring generated over a number field, has no input here. No operator of order one has algebraic values on the lattice. The operator that does, $\partial_z\partial_w$, with $\partial_z\partial_w F = u\bar{u}F = \rho F$, is exactly where the candidate’s relation enters, and it is not a derivation.

Proposition 5.7 (Kronecker factorisation). *Let u be a candidate and $\alpha = e^u$. For $0 \leq a, b, m, n \leq N$ the matrix $M = (\exp(aum + b\bar{u}n))$, with rows indexed by (a, b) and columns by (m, n) , is the Kronecker product $A \otimes B$ of $A = (\alpha^{am})$ and $B = (\bar{\alpha}^{bn})$. Hence*

$$\det M = (\det A)^{N+1}(\det B)^{N+1} \neq 0.$$

Proof. $\exp(aum + b\bar{u}n) = \alpha^{am}\bar{\alpha}^{bn} = A_{a,m}B_{b,n}$, which is the Kronecker product, and $\det(A \otimes B) = (\det A)^{N+1}(\det B)^{N+1}$. Each factor is a Vandermonde determinant, in the nodes α^a and $\bar{\alpha}^b$ respectively, and the nodes are pairwise distinct because $|\alpha| = e^{\operatorname{Re} u} \neq 1$, a candidate having $\operatorname{Re} u \neq 0$. $\square$

So the interpolation determinant of the natural system splits into two independent one-variable determinants, and neither ρ nor the relation $u\bar{u} = \rho$ appears in it. Schwarz’ lemma for Cartesian products [8, Proposition 4.7] asks for vanishing on a full simplex of derivatives, which throws the method back on the values; and the values, by Proposition 5.7, see nothing of the candidate’s arithmetic.

6 Exclusions outside the hull

Section 5 showed that nothing inside $W_u = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u + \overline{\mathbb{Q}}\bar{u}$ decides the question: the hull a candidate certifies out of itself is three-dimensional, and the templates that would refute it need four dimensions. The way past that is to leave the hull — to adjoin one further number and ask what it would have to be. This section collects what that yields. Everything here is a necessary

condition on a hypothetical counterexample; none of it is an emptiness proof, and the final subsection says where the method stops.

Throughout, u is a candidate, $\rho = |u|^2 \in \overline{\mathbb{Q}}^\times$, and $\bar{u} = \rho/u$, so that $u^{-1} = \bar{u}/\rho$ lies in $\tilde{\mathcal{L}}$ along with u and 1.

Algebraic lines, and distance to an algebraic point

Theorem 6.1. *Let $\lambda \in \mathcal{L}$ with $\operatorname{Re} \lambda \neq 0$ and $\operatorname{Im} \lambda \neq 0$. There are no $B \in \overline{\mathbb{Q}}^\times$ and $C \in \overline{\mathbb{Q}}$ with*

$$B\lambda + \bar{B}\bar{\lambda} + C = 0.$$

Proof. Off both axes, λ and $\bar{\lambda}$ are $\mathbb{Q}$ -linearly independent: taking real and imaginary parts of $p\lambda + q\bar{\lambda} = 0$ with $p, q \in \mathbb{Q}$ gives $(p+q)\operatorname{Re} \lambda = 0$ and $(p-q)\operatorname{Im} \lambda = 0$, whence $p = q = 0$. Both lie in $\mathcal{L}$, since $e^\lambda = e^{\bar{\lambda}}$ is algebraic. Baker's theorem in its inhomogeneous form then makes the non-zero $\overline{\mathbb{Q}}$ -combination $B\lambda + \bar{B}\bar{\lambda}$ transcendental, so it cannot equal the algebraic number $-C$. $\square$

The condition has a geometric reading. Writing $B = B_1 + iB_2$ and $\lambda = x + iy$, the relation is $2(B_1x - B_2y) + C = 0$: an affine line with real algebraic coefficients, not both zero. So Theorem 6.1 says a logarithm off the axes lies on no such line. For a candidate the same conclusion is available with less input, and it is worth recording which:

Remark 6.2. For a candidate $u = x + iy$, no relation $Ax + By = C$ with $A, B, C \in \overline{\mathbb{Q}} \cap \mathbb{R}$ and $(A, B) \neq (0, 0)$ holds, by Hermite–Lindemann alone. Such a line meets the circle $X^2 + Y^2 = \rho$ only in points with algebraic coordinates — solve the linear equation for one coordinate and substitute, the resulting quadratic having non-zero leading coefficient because A, B are real — and u is transcendental. No linear forms in logarithms are needed. This form now has a machine-checked proof of its own; Theorem 6.1 remains the stronger statement, dropping the hypothesis on $|u|$ at the cost of Baker's theorem. It is also the degree-one case of Proposition 6.14 below: a non-zero polynomial of degree at most one cannot be divisible by $X^2 + Y^2 - \rho$.

Corollary 6.3. *For every $a \in \overline{\mathbb{Q}}^\times$, $|u - a| \notin \overline{\mathbb{Q}}$. Equivalently, a candidate lies on no circle of algebraic radius about a non-zero algebraic centre. This statement is now machine-checked.*

Proof. If $|u - a|$ were algebraic then so would be $2\operatorname{Re}(\bar{a}u) = |u|^2 + |a|^2 - |u - a|^2$, which is a forbidden line by Remark 6.2. The exclusion of $a = 0$ is not cosmetic: $|u|$ is algebraic by hypothesis. $\square$

The multiplier module

The next statement determines, completely, which numbers can multiply a candidate back into $\tilde{\mathcal{L}}$. Its input is Roy's strong six exponentials theorem: for $\overline{\mathbb{Q}}$ -linearly independent x_1, x_2 and $\overline{\mathbb{Q}}$ -linearly independent y_1, y_2, y_3 , at least one of the six products $x_i y_j$ lies outside $\tilde{\mathcal{L}}$.

Theorem 6.4. *Assume the strong six exponentials theorem. For every candidate u ,*

$$\{z \in \tilde{\mathcal{L}} : uz \in \tilde{\mathcal{L}}\} = \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u^{-1}. \quad (\text{M})$$

Proof. For the inclusion $\supseteq$: 1 and u^{-1} lie in $\tilde{\mathcal{L}}$, and for $z = a + bu^{-1}$ with $a, b \in \overline{\mathbb{Q}}$ one has $uz = au + b \in \tilde{\mathcal{L}}$.

For $\subseteq$, suppose $z \in \tilde{\mathcal{L}}$, $uz \in \tilde{\mathcal{L}}$ and $z \notin \overline{\mathbb{Q}} + \overline{\mathbb{Q}}u^{-1}$. Since u is transcendental, 1, u are $\overline{\mathbb{Q}}$ -linearly independent, and so are $1, u^{-1}$; the assumption on z extends the second pair to an $\overline{\mathbb{Q}}$ -linearly independent triple $z, 1, u^{-1}$. The six products of $(1, u)$ with $(z, 1, u^{-1})$ are

$$z, \quad 1, \quad u^{-1}, \quad uz, \quad u, \quad u \cdot u^{-1} = 1,$$

all of them in $\tilde{\mathcal{L}}$. This contradicts the strong six exponentials theorem. $\square$

The force of (M) is that the right-hand side is closed under nothing useful. Two instances:

Corollary 6.5. *Assume the strong six exponentials theorem. For every candidate u and all $a \in \overline{\mathbb{Q}}^\times$, $b, c \in \overline{\mathbb{Q}}$,*

$$u^2 \notin \tilde{\mathcal{L}}, \quad \text{hence} \quad au^2 + bu + c \notin \tilde{\mathcal{L}}.$$

In particular e^{au^2+bu+c} is transcendental.

Proof. Take $z = u$ in (M). If $u^2 \in \tilde{\mathcal{L}}$ then $u = a' + b'u^{-1}$ for some $a', b' \in \overline{\mathbb{Q}}$, so $u^2 - a'u - b' = 0$ and u is algebraic. The second form follows because $\tilde{\mathcal{L}}$ contains u and $\overline{\mathbb{Q}}$ and is an $\overline{\mathbb{Q}}$ -subspace. $\square$

Corollary 6.6. *Assume the strong six exponentials theorem. For every candidate u and every $a \in \overline{\mathbb{Q}}^\times$,*

$$\frac{1}{u-a} \notin \tilde{\mathcal{L}}.$$

Proof. Put $z = (u-a)^{-1}$, a legitimate test element because $uz = 1 + az$, so $uz \in \tilde{\mathcal{L}}$ as soon as z does. Then (M) gives $z = c + du^{-1}$ with $c, d \in \overline{\mathbb{Q}}$, and clearing denominators,

$$cu^2 + (d - ca - 1)u - da = 0.$$

The coefficients are algebraic and not all zero: $c = 0$ forces $d = 1$, and then the constant term is $-a \neq 0$. So u is algebraic, a contradiction. $\square$

The contrast is the point. A candidate's reciprocal $1/u$ necessarily lies in $\tilde{\mathcal{L}}$, while no algebraically shifted reciprocal does.

Commensurable pairs, unconditionally

Corollary 6.5 rests on Roy's theorem, which has no formal proof. Part of it survives on the four exponentials theorem in transcendence degree one alone, through a statement about pairs of logarithms.

Theorem 6.7. *Let $u, v \in \mathcal{L} \setminus \{0\}$ with $|u|^2/|v|^2 \in \mathbb{Q}$, and suppose $\text{trdeg}_{\mathbb{Q}} \mathbb{Q}(u, \bar{u}, v, \bar{v}) \leq 1$. Then $v \in \mathbb{Q}u$ or $v \in \mathbb{Q}\bar{u}$.*

Proof. With $c = |u|^2/|v|^2$, the matrix $\begin{pmatrix} u & v \\ c\bar{v} & \bar{u} \end{pmatrix}$ has determinant $|u|^2 - c|v|^2 = 0$ and non-zero entries in $\mathcal{L}$. A column relation gives $v \in \mathbb{Q}u$, a row relation $v \in \mathbb{Q}\bar{u}$. $\square$

The hypotheses are satisfiable, by $u = \log 2$ and $v = \log 4$, and both alternatives occur. For a candidate, $\bar{u} = \rho/u$, so the transcendence-degree hypothesis holds for every v algebraic over $\mathbb{Q}(u)$.

Corollary 6.8. *Let u be a candidate, $k \geq 2$, and $c \in \overline{\mathbb{Q}}^\times$ with $|c|^2\rho^{k-1} \in \mathbb{Q}$. Then e^{cu^k} is transcendental; in particular $e^{u^k/|u|^{k-1}}$ is, for every $k \geq 2$.*

Proof. If $v = cu^k \in \mathcal{L}$, then $|v|^2 = (|c|^2\rho^{k-1})\rho$, and Theorem 6.7 gives $cu^k = qu$ or $cu^k = q\rho/u$ with $q \in \mathbb{Q}$. Either way a positive power of u is algebraic, so u is algebraic, against Hermite–Lindemann. $\square$

Negative exponents follow by applying the corollary to $\bar{u}$, since $cu^{-k} = (c/\rho^k)\bar{u}^k$. The same matrix argument, with $\begin{pmatrix} u & m \\ l & \bar{u} \end{pmatrix}$ for $l = pu + q\bar{u}$ and $m = \rho/l$, shows that $e^{\rho/(pu+q\bar{u})}$ is transcendental for all non-zero rationals p, q . In particular the harmonic mean $\rho/\text{Re } u$ of u and $\bar{u}$ is not in $\mathcal{L}$.

Two candidates, and the six exponentials theorem

Theorem 5.2 says the six exponentials theorem cannot refute one candidate: its certificate is three-dimensional. Two candidates whose squared moduli have rational ratio supply the missing dimension.

Theorem 6.9. *Let $u, v \in \mathcal{L} \setminus \{0\}$ with $|u|^2/|v|^2 = c \in \mathbb{Q}$ and $v \notin \mathbb{Q}u \cup \mathbb{Q}\bar{u}$. If $w \in \mathcal{L}$ and $uw/v \in \mathcal{L}$, then $w \in \mathbb{Q}v + \mathbb{Q}\bar{u}$.*

Proof. Put $m = u/v \notin \mathbb{Q}$. Then $mv = u$ and $m\bar{u} = c\bar{v}$ lie in $\mathcal{L}$. If $w \notin \mathbb{Q}v + \mathbb{Q}\bar{u}$, the six products of $x = (1, m)$ and $y = (v, \bar{u}, w)$ all lie in $\mathcal{L}$, with x and y both $\mathbb{Q}$ -independent, against the six exponentials theorem. $\square$

Two lemmas turn this into a statement about candidates. If p, q are candidates and $\operatorname{Re}(pq) \in \overline{\mathbb{Q}}$, then $q \in \mathbb{Q}\bar{p}$: indeed pq is then algebraic, since $|pq|^2$ is, and Gel'fond–Schneider applied to $q = (q/\bar{p})\bar{p}$ forces $q/\bar{p} \in \mathbb{Q}$. And if $u \in \mathcal{L}$ lies off both axes and $w, uw/\bar{u} \in \mathcal{L}$, then $a|u|^2 + b|uw| + c|\bar{u}w| = 0$ for rationals a, b, c not all zero. That lemma is six exponentials with $x = (1, u/\bar{u})$ and $y = (\bar{u}, w, (u/\bar{u})w)$.

Theorem 6.10. *Let u, v, w be candidates with $|u|^2/|v|^2 \in \mathbb{Q}$. If $e^{uw/v} \in \overline{\mathbb{Q}}$, then $v \in \mathbb{Q}u$, $w \in \mathbb{Q}v$ or $w \in \mathbb{Q}\bar{u}$.*

Proof. If $v \in \mathbb{Q}\bar{u}$, the second lemma makes $\operatorname{Re}(uw)$ algebraic, and the first gives $w \in \mathbb{Q}\bar{u}$. Otherwise, if $v \notin \mathbb{Q}u$, Theorem 6.9 gives $w = av + b\bar{u}$, and $|w|^2 = a^2|v|^2 + b^2|u|^2 + 2ab\operatorname{Re}(uv)$. If $ab \neq 0$ this makes $\operatorname{Re}(uv)$ algebraic, and the first lemma puts $v \in \mathbb{Q}\bar{u}$. $\square$

Corollary 6.11. *For real algebraic $r > 0$ let $\Phi_r \subset \mathbb{R}/\pi\mathbb{Z}$ be the set of arguments of the candidates of modulus r . If $\varphi_1 + \varphi_2 = \varphi_3 + \varphi_4 \neq 0$ with all $\varphi_j \in \Phi_r$, then $\{\varphi_1, \varphi_2\} = \{\varphi_3, \varphi_4\}$. In particular Φ_r contains no three-term arithmetic progression.*

Proof. The relation gives $u_1u_2 = \pm u_3u_4$ for candidates u_j of modulus r , so $e^{u_1u_2/u_3} \in \overline{\mathbb{Q}}$, and Theorem 6.10 applies. The third alternative, $u_2 \in \mathbb{Q}\bar{u}_1$, means $\varphi_1 + \varphi_2 = 0$. $\square$

So the arguments of candidates on a circle form a Sidon set, up to conjugation. Two candidates of commensurable moduli that are not trivially related generate transcendence degree two (Theorem 6.7), where the four exponentials theorem in degree one is silent; this is the first constraint of any kind on such families.

The same two tools reach an exponent that Gel'fond–Schneider cannot. For every candidate u ,

$$(-1)^{u/\bar{u}} := e^{i\pi u/\bar{u}} \text{ is transcendental.}$$

If it were algebraic, the second lemma with $w = i\pi$ would give $\pi \operatorname{Im} u \in \mathbb{Q}\rho$, making u algebraic over $\mathbb{Q}(\pi)$; and then the four exponentials theorem in transcendence degree one, applied to $\begin{pmatrix} i\pi u/\bar{u} & i\pi \\ u & \bar{u} \end{pmatrix}$, gives a contradiction, since neither $i\pi \in \mathbb{Q}\bar{u}$ nor a rational relation between u and $\bar{u}$ is possible.

Saturation at one logarithm

Theorem 6.12. *Let u be a candidate and $\ell \in \mathcal{L}$. If $u \in \overline{\mathbb{Q}} + \overline{\mathbb{Q}}\ell$ then $u \in \mathbb{Q}\ell$.*

Proof. Write $u = a + b\ell$ with $a, b \in \overline{\mathbb{Q}}$. If u and ℓ are $\mathbb{Q}$ -linearly independent, Baker's inhomogeneous theorem makes $u - b\ell$ transcendental, and it equals a . So they are $\mathbb{Q}$ -dependent: $pu + q\ell = 0$ with $p, q \in \mathbb{Q}$ not both zero. Here $p \neq 0$, since $p = 0$ would give $q\ell = 0$ and hence $\ell = 0$, making $u = a$ algebraic against Hermite–Lindemann. Thus $u = (-q/p)\ell$. $\square$

So inside an algebraic line spanned by one logarithm, the elements of $\mathcal{L}$ are exactly the *rational* multiples of that logarithm: the algebraic span collapses to the rational one. The instance $\ell = i\pi$ closes off a plane one might have expected to have to examine.

Corollary 6.13. *No candidate has the form $a + b\pi$ with $a, b \in \overline{\mathbb{Q}}$.*

Proof. $i\pi \in \mathcal{L}$, and $a + b\pi = a + (-ib)(i\pi)$ with $-ib$ algebraic. By Theorem 6.12, $u = r i\pi$ for some $r \in \mathbb{Q}$, necessarily non-zero; then $|u| = |r|\pi$ is transcendental, contradicting the candidate's algebraic modulus. $\square$

The boundary of this argument should be stated explicitly, because it is easy to over-read. Theorem 6.12 handles one logarithm at a time, with the candidate itself as the second. It says nothing about a combination of two independent logarithms, and in particular it does not touch $\log 2 + i\pi$, a logarithm of -2 and the smallest open test case of §3. The distance between the two situations is the whole difficulty.

Generic on its own circle

One further exclusion belongs with these, and it is the strongest of them. It was the last statement in this note to acquire a machine-checked proof, and the route the formalisation took is not the one given below.

Proposition 6.14. *Let $u = x + iy$ be a candidate and $\rho = |u|^2$. For every $P \in \overline{\mathbb{Q}}[X, Y]$,*

$$P(x, y) = 0 \iff X^2 + Y^2 - \rho \text{ divides } P.$$

Proof. The implication $\Leftarrow$ is immediate from $x^2 + y^2 = \rho$. For $\Rightarrow$, make the invertible linear change of variables $U = X + iY$, $V = X - iY$ over $\overline{\mathbb{Q}}$, under which $X^2 + Y^2$ becomes UV ; let $Q \in \overline{\mathbb{Q}}[U, V]$ be the image of P , so that $Q(u, \bar{u}) = P(x, y) = 0$ and $\bar{u} = \rho/u$. Modulo $UV - \rho$ the ring $\overline{\mathbb{Q}}[U, V]$ becomes $\overline{\mathbb{Q}}[U, U^{-1}]$, and the residue of Q is a Laurent polynomial vanishing at u . Since u is transcendental, a non-zero Laurent polynomial over $\overline{\mathbb{Q}}$ cannot vanish at u : clearing denominators would give an algebraic relation for u . Hence the residue is zero, $Q \in (UV - \rho)$, and $P \in (X^2 + Y^2 - \rho)$. $\square$

A candidate is therefore algebraically generic on its own circle: it lies on no algebraic plane curve that does not contain that entire circle. It lies on no parabola, and on no ellipse or hyperbola other than through the circle appearing as a component. What the proposition does not do is eliminate the circle itself, which is precisely where the difficulty has been all along — so the statement is sharp in form and empty in consequence for the main question.

The formal proof takes a different route from the one above, and the difference is instructive. Rather than passing to the coordinate ring of the conic, it reads P in $\overline{\mathbb{Q}}[X][Y]$ and divides by the monic quadratic $Y^2 + (X^2 - \rho)$, leaving a remainder $A(X)Y + B(X)$ which must vanish at (x, y) . Two facts finish it: x and y are each transcendental — if either were algebraic, $x^2 + y^2 = \rho$ would make the other algebraic and hence u algebraic — and $y \notin \overline{\mathbb{Q}}(x)$, because $\rho - X^2$ is squarefree of degree two and so not a square in $\overline{\mathbb{Q}}(X)$. The change of variables is the shorter argument on paper; the division is the one a proof assistant prefers, because it needs no quotient ring.

7 What would have to be proved

Three statements, in increasing strength, each of which settles Conjecture 1.1.

- (1) The single relation of Theorem 3.5, for the branch it governs: for real $t \neq 0$ with e^t algebraic, $t^2 + \pi^2$ is transcendental. Smallest open instance: is $\sqrt{(\log 2)^2 + \pi^2}$ algebraic?

- (2) The rank inequality of §4(a) over $(\overline{\mathbb{Q}}, \mathbb{C}, \tilde{\mathcal{L}})$, together with an inhomogeneous extension of Proposition 12.25. The first component is under active study; the second is not.
- (3) The strong four exponentials conjecture, which settles the whole question at once by Theorem 2.1.

Nothing here brings any of the three closer. What the development does establish is where the question is not: not in a better matrix, by Theorem 4.2; not in an algebraic invariant of the candidate's hull, by Theorem 5.1; not in the six exponentials theorem, by Theorem 5.2; and not, on one whole branch, in further case analysis, by Proposition 5.5.

Where the formal development stops

The case analysis behind this note has been carried to its end. Every branch of Conjecture 1.1 is either closed by a machine-checked proof or reduced, by a machine-checked reduction, to one of exactly two statements, and nothing else in it is open:

- (i) the real half of (S): for real algebraic $\gamma \neq 0$, $e^{-i\gamma/\pi}$ is transcendental;
- (ii) the statement (NT) of Proposition 5.5, which is equivalent to the conjecture itself.

Both follow from Conjecture 1.2. Neither is a formalisation problem. Version 1.0 of this note listed a third, the imaginary half of (S): for real algebraic $\beta \neq 0$, $e^{\beta/\pi}$ is transcendental. It remains open, but the development no longer depends on it, since both branches that reach (S) produce a real γ . By Theorem 3.12, it and the real half cannot both fail.

It is worth saying why the four exponentials theorem in transcendence degree one, now proved, does not reach them, since each lives in transcendence degree one: under a counterexample to (S) the numbers λ and $i\pi$ are algebraically dependent, $\lambda = \gamma/(i\pi)$, and a candidate's u and $\bar{u}$ are algebraically dependent because $\bar{u} = \rho/u$. The obstruction is the input, not the degree. The configurations that Theorems 2.1 and 3.10 use have products $1, \bar{u}, u, \rho$ and $1, i\pi, \lambda, \gamma$, and in each two of the four are non-zero algebraic numbers. A non-zero algebraic number is never in $\mathcal{L}$, by Hermite–Lindemann. The four exponentials theorem, and the four exponentials conjecture above it, ask for their products to be logarithms of algebraic numbers; only the strong form, whose products are allowed to range over $\tilde{\mathcal{L}}$, accepts these. That distinction, $\mathcal{L}$ against $\tilde{\mathcal{L}}$, is the whole distance between what is proved and what is needed. Theorem 5.3 and Proposition 5.4 make it precise. On the certified logarithms the four exponentials conjecture is already a theorem, so it can give nothing. What separates $\tilde{\mathcal{L}}$ from $\mathcal{L}$ here is the constant term, not the algebraic coefficients.

References

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- [2] G. Diaz, *Produits et quotients de combinaisons linéaires de logarithmes de nombres algébriques : conjectures et résultats partiels*, J. Théor. Nombres Bordeaux **19** (2007), 373–391. It combines Roy's strong six exponentials theorem with complex conjugation to obtain cases of the strong four exponentials conjecture; Roy's theorem is not formalised, so those results are not used here.
- [3] M. Karataarakis and F. Wiedijk, *A formalization of the Gelfond-Schneider theorem*, arXiv:2603.24823 (2026).
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- [5] W. D. Brownawell, *The algebraic independence of certain numbers related to the exponential function*, J. Number Theory **6** (1974), 22–31.
- [6] M. Waldschmidt, *Solution du huitième problème de Schneider*, J. Number Theory **5** (1973), 191–202.
- [7] M. Waldschmidt, *Indépendance algébrique des valeurs de la fonction exponentielle*, Bull. Soc. Math. France **99** (1971), 285–304.
- [8] M. Waldschmidt, *Diophantine Approximation on Linear Algebraic Groups*, Grundlehren der mathematischen Wissenschaften **326**, Springer, 2000.
- [9] S. Dasgupta and M. Kakde, *On ranks of matrices of logarithms*, preprint.

A Machine-checked certificates

Every numbered statement above but one has been checked in a formal development, in Lean 4 against a pinned revision of its mathematical library. The table records where each one lives, so that any claim in this note can be verified rather than taken on trust. The environment is stated here once and nowhere else.

The proofs exist in two places, and the difference matters to a reader who wants to check them. Each identifier below names a node on a collaborative proving platform, where the statement, the submitted proof and its verdict are public. Every proved statement is also mirrored in a self-contained Lean project at

<https://github.com/carlok/diaz-modulus-lean>

which builds from source in one command, depends on no platform, and assumes nothing beyond Lean’s own three axioms. Where the two differ in naming, the project keeps the platform’s spelling.

This is version 1.3 of the note; version 1.0 of 21 September 2026 is the first stable release. Every row of the table below was checked against the platform by script on 24 September 2026, and the mirrored project held all 200 proved results of the development at that date.

The 1973–74 theorem’s row stands for a development rather than a single proof: the criterion of Waldschmidt 1971, the zero count for exponential polynomials, the construction of 1973 with its auxiliary function, extrapolation and norm step, and the assembly above them. They are separate nodes under the `FourExp` namespace on the platform, and separate modules in the project.

Two further statements, not numbered above because the note does not use them directly, are proved alongside Proposition 6.14 and belong to its argument: the real and imaginary parts of a candidate are each transcendental (`DiazModulus.candidate_re_transcendental`, `DiazModulus.candidate_im_transcendental`).

A **Proved** row means the statement itself has a machine-checked proof. An **Open** row means the statement is recorded formally but not proved; every such row here is an open conjecture. A **Not formalised** row means what it says: the statement is proved in this note and has no formal counterpart at all. Proposition 5.5 is flagged that way, and its row needs a word: the identifier given is the statement (NT), which is recorded formally and is Open; the *equivalence* proved in Proposition 5.5 is what has no formal counterpart.

Result	Status	Identifier
Conj. 1.1	Open (conjecture)	<code>DiazModulus.diaz_modulus_conjecture</code>
Thm 2.1	Proved	<code>DiazModulus.diaz_of_sfe</code>
Rem. 2.2	Proved	<code>DiazModulus.diaz_of_schanuel</code>

Result	Status	Identifier
Thm 3.1	Proved	Diaz.real_quantisation
Prop. 3.2	Proved	Diaz.quantisation_orbit_iff_re_ne_zero
Thm 3.3	Proved	Diaz.fibre_at_most_two
Cor. 3.4	Proved	Diaz.q_translate_unique
Thm 3.5	Proved	DiazModulus.leaf_iff_one
Thm 3.6	Proved	DiazModulus.diaz_number_forces_transcendence
Cor. 3.7	Proved	DiazModulus.log_two_diaz_or_transcendental
Thm 4.1	Proved	Diaz.no_vanishing_coeff
Thm 4.2	Proved	Diaz.det_pencil_eq_conic
Prop. 4.3	Proved	Diaz.roy_conic_implies_empty
Thm 5.1	Proved	Diaz.candidate_indistinguishable_by_coeff
Thm 5.2	Proved	DiazModulus.sixExponentials_cannot_refute_candidate
Thm 5.3	Proved	DiazModulus.generic_conj_pair_no_quadratic_relation, DiazModulus.generic_conj_pair_four_exp_barrier
Prop. 5.4	Proved	DiazModulus.candidate_nongeneric_four_exp_barrier
§5, the barrier for (S)	Proved	DiazModulus.recip_pi_log_four_exp_barrier
§5, 2×2 matrices of linear forms	Proved	DiazModulus.det_zero_linear_forms_rank_one
Prop. 5.5	Not formalised	DiazModulus.norm_transcendental_of_generic_conj_pair (Open)
§5, 1973–74 thm	Proved	DiazModulus.four_exponentials_trdeg_one
Prop. 3.8	Proved	DiazModulus.recip_pi_log_of_period_aligned
Conj. 3.9	Open (conjecture)	DiazModulus.recip_pi_not_log
Thm 3.10	Proved	DiazModulus.recip_pi_not_log_of_sfe
Prop. 3.11	Proved	DiazModulus.recip_pi_exp_value_not_root_of_unity
Thm 3.12(a)	Proved	DiazModulus.recip_pi_log_rational_line
Thm 3.12(b)	Proved	DiazModulus.recip_pi_log_on_axis
Thm 3.12(c)	Proved	DiazModulus.recip_pi_not_log_real_or_imag
§3, $\pi \log 2$ or $\pi \log 3$	Proved	DiazModulus.pi_log_two_or_pi_log_three_transcendental
§3, partners of $e^{i\gamma/\pi}$	Proved	DiazModulus.recip_pi_or_pi_cube
§3, $e^{i/\pi}$ or $e^{i\pi^3}$	Proved	DiazModulus.exp_i_div_pi_or_exp_i_pi_cube_transcendental
§3, real half of (S)	Open (conjecture)	DiazModulus.recip_pi_not_log_real_gamma
§3, imaginary half of (S)	Open (conjecture)	DiazModulus.recip_pi_not_log_imag_gamma
Prop. 5.6	Proved	DiazModulus.no_first_order_arithmetic_operator
Prop. 5.7	Proved	DiazModulus.kronecker_factorisation
Prop. 4.4	Proved	DiazModulus.geometric_triple_not_logs
Cor. 4.5	Proved	DiazModulus.exp_pi_sq_or_exp_i_pi_cube_transcendental
§5, 2^π , 3^π or 5^π	Proved	DiazModulus.two_three_five_pow_pi_transcendental
§5, $e^{\lambda^2/\mu}$ or $e^{\mu^2/\lambda}$	Proved	DiazModulus.log_square_duality
§5, $2^{\log_3 2}$ or $3^{\log_2 3}$	Proved	DiazModulus.two_pow_log_three_or_three_pow_log_two

Result	Status	Identifier
Thm 6.1	Proved	DiazModulus.no_algebraic_generalized_line
Rem. 6.2	Proved	DiazModulus.candidate_no_real_algebraic_line
Cor. 6.3	Proved	DiazModulus.candidate_distance_transcendental
Thm 6.4	Proved	DiazModulus.candidate_multiplier_module
Cor. 6.5	Proved	DiazModulus.candidate_multiplier_module
Cor. 6.6	Proved	DiazModulus.candidate_multiplier_module
Thm 6.7	Proved	DiazModulus.log_pair_rigid_of_trdeg_one
Cor. 6.8	Proved	DiazModulus.candidate_monomial_not_log
§6, harmonic mean	Proved	DiazModulus.candidate_harmonic_not_log
Thm 6.9	Proved	DiazModulus.log_ratio_multipliers
§6, the two lemmas	Proved	DiazModulus.candidate_conj_product_rational, DiazModulus.conj_ratio_multiplier_relation
Thm 6.10	Proved	DiazModulus.candidate_quotient_rigid
Cor. 6.11	Proved	DiazModulus.candidate_product_relation_trivial
§6, $(-1)^{u/\bar{u}}$	Proved	DiazModulus.candidate_neg_one_pow_ratio
Thm 6.12	Proved	DiazModulus.candidate_one_log_saturation
Cor. 6.13	Proved	DiazModulus.candidate_one_log_saturation
Prop. 6.14	Proved	DiazModulus.candidate_vanishing_ideal
Hermite–Lindemann	Proved	DiazModulus.hermite_lindemann_holds

Two entries need a word. Hermite–Lindemann is used freely in the text as classical; it is listed because the library used does not contain it, so it had to be established rather than cited. The same holds for Gel’fond–Schneider, which is proved on the platform in another mission, from the formalisation of Karatarakis and Wiedijk, and is mirrored in the project. And the statement (NT) of Proposition 5.5 was contributed by another participant in the same development, independently of the work reported here; the equivalence with Conjecture 1.1 is what is established above.

Some results in the development are not represented here at all. Roughly a third of it consists of closure lemmas, coercions and bookkeeping with no content at this level of exposition, and a further group restates in one set of coordinates what another states in a different one. Neither belongs in a paper, and neither is listed.